

SAENZ PENA, Argentina

WMO: 871480

Lat: 26.7464S

Lon: 60.4867W

Elev: 305

StdP: 14.53

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.1	38.0	26.6	20.8	47.8	30.1	24.4	48.8	27.6	73.1	26.1	77.5	1.6	230	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.6	100.5	76.7	98.1	76.8	95.8	76.2	81.2	93.0	80.0	91.6	79.1	90.1	12.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.3	148.4	87.2	77.1	142.9	85.8	76.1	138.0	84.8	45.4	93.1	44.0	92.0	42.9	90.3	93.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.2	20.5	17.5	DB	26.5	105.9	4.1	2.7	23.5	107.8	21.1	109.4	18.8	110.9	15.8	112.9	(4)
(5)			WB	26.6	83.7	4.5	2.6	23.3	85.6	20.7	87.2	18.2	88.6	14.9	90.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	71.3	81.3	79.8	76.1	71.2	64.8	61.0	60.4	63.8	68.3	73.3	75.9	79.7	(6)	
	DBStd	10.66	4.81	5.93	5.80	7.45	8.44	9.13	10.12	10.59	9.61	7.65	6.15	5.69	(7)	
	HDD50	36	0	0	0	0	2	9	17	7	1	0	0	0	(8)	
	HDD65	790	0	0	2	26	111	187	219	158	71	14	2	0	(9)	
	CDD50	7801	971	835	809	637	462	341	340	435	549	722	778	922	(10)	
	CDD65	3078	506	415	346	212	106	68	76	121	169	271	331	457	(11)	
	CDH74	32350	5951	4510	3264	1748	744	403	626	1396	1930	2852	3584	5342	(12)	
	CDH80	15221	3067	2206	1442	656	225	93	213	656	917	1342	1680	2724	(13)	
(14) Wind	WSAvg	7.0	6.2	6.2	5.6	5.9	5.9	6.5	7.1	8.3	8.8	8.8	7.5	6.7	(14)	
(15) Precipitation	PrecAvg	41.5	5.8	5.2	5.7	5.6	2.1	1.0	0.8	0.9	1.6	3.0	5.1	5.1	(15)	
	PrecMax	68.1	15.8	11.5	15.6	14.3	7.3	3.9	3.6	2.3	5.1	9.5	9.5	12.0	(16)	
	PrecMin	25.3	0.7	0.9	0.9	0.8	0.0	0.0	0.0	0.0	0.0	0.7	0.6	0.7	(17)	
	PrecStd	10.0	3.1	3.1	3.2	3.3	1.7	1.0	0.8	0.8	1.3	1.9	2.4	2.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.5	101.5	99.6	94.0	90.8	86.1	89.8	97.0	101.4	102.9	101.6	102.7	(19)	
		MCWB	77.8	77.9	76.1	75.8	75.1	72.1	68.7	72.4	73.0	75.9	77.3	77.1	(20)	
	2%	DB	98.6	98.0	95.2	90.8	85.8	82.9	85.8	92.8	96.0	97.4	96.7	98.6	(21)	
		MCWB	77.7	78.0	76.0	75.7	72.6	71.2	69.3	70.7	71.6	76.1	76.0	77.7	(22)	
	5%	DB	96.0	94.8	91.5	87.4	81.5	79.2	81.5	88.0	90.3	92.2	92.9	95.6	(23)	
		MCWB	77.2	77.5	75.4	74.1	71.4	69.9	67.9	69.8	70.1	73.6	74.4	76.8	(24)	
	10%	DB	93.3	91.4	87.9	83.8	77.6	74.6	76.2	82.0	84.7	87.5	89.2	92.6	(25)	
		MCWB	76.8	76.6	74.5	72.9	69.6	67.8	65.4	67.1	67.7	71.3	72.8	75.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.7	82.7	81.5	79.8	77.3	74.6	73.5	75.1	76.4	79.8	81.3	81.9	(27)	
		MCDB	93.6	93.5	92.0	89.6	85.4	81.9	84.1	92.4	93.6	93.8	94.6	94.9	(28)	
	2%	WB	80.9	80.8	79.3	78.1	75.0	72.5	70.8	72.3	74.1	77.6	79.0	80.2	(29)	
		MCDB	92.5	92.1	89.1	86.6	83.0	79.7	81.6	88.8	88.9	92.2	91.5	92.6	(30)	
	5%	WB	79.6	79.2	77.7	76.5	73.3	70.7	68.8	70.0	72.0	75.7	77.1	79.1	(31)	
		MCDB	90.8	90.1	87.0	84.4	79.6	77.5	78.7	85.6	86.1	89.4	88.8	91.0	(32)	
	10%	WB	78.5	77.9	76.2	74.6	71.1	68.6	66.7	67.6	69.9	73.7	75.3	77.9	(33)	
		MCDB	88.9	87.9	84.6	81.1	76.1	74.5	75.7	80.1	82.3	85.3	85.3	89.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	20.6	19.3	19.8	18.4	18.8	18.7	22.0	24.2	23.9	21.1	21.7	20.9	(35)	
		MCDBR	24.6	23.3	24.8	22.2	21.4	20.1	24.6	29.0	29.6	27.5	27.2	25.0	(36)	
	5% WB	MCWBR	8.1	7.6	9.4	9.2	9.9	9.8	10.8	11.6	11.9	10.6	10.5	8.2	(37)	
		MCWBR	19.8	19.3	19.7	17.8	16.8	17.8	22.2	27.3	24.7	23.7	21.7	20.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.408	0.405	0.393	0.395	0.369	0.375	0.370	0.462	0.539	0.476	0.403	0.408	(40)	
		taud	2.391	2.407	2.429	2.391	2.424	2.397	2.388	2.090	1.889	2.138	2.363	2.379	(41)	
	Ebn at Noon	Ebn at Noon	297	293	286	270	263	253	259	244	240	269	296	298	(42)	
		Edn at Noon	41	39	36	35	30	30	31	46	62	51	42	42	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2141	1946	1675	1282	1025	856	1030	1282	1530	1765	2069	2122	(44)	
	RadStd	RadStd	162	128	112	139	125	86	81	116	178	173	146	149	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page